Form C

# Core Focus on Proportions & Probability

## Block 2 Review ~ Proportions and Similarity

**ANSWER KEY**

Name _____ Period _____ Date _____

## Part I – Selected Response

**1.** What is the value of $x$ in the proportion $\frac{6}{15} = \frac{x}{25}$?

A. 8

B. 10

C. 12

D. 16

B (15x = 6 · 25 = 150, so x = 10)

**2.** Mrs. Lopez paid $40.50 for 6 museum tickets. How much would 10 museum tickets cost?

A. $44.50

B. $60.75

C. $67.50

D. $81.00

C ($40.50 ÷ 6 = $6.75 each; 10 × $6.75 = $67.50)

**3.** Theo reads 5 pages in 12 minutes. How long would it take him to read 35 pages at this pace?

A. 42 minutes

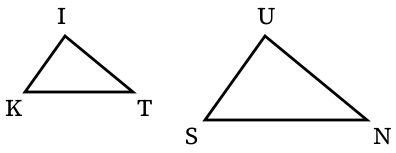
B. $14\frac{7}{12}$ minutes

C. 84 minutes

D. 60 minutes

C (5/12 = 35/x → 5x = 420 → x = 84 minutes)

**4.** △KIT is similar to △SUN. Which side of △KIT corresponds to $\overline{UN}$ in △SUN?




A. $\overline{KI}$

B. $\overline{IT}$

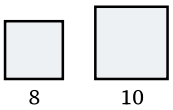
C. $\overline{TK}$

D. $\overline{KT}$

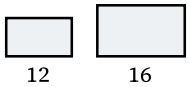
B (U ↔ I and N ↔ T, so UN ↔ IT)

**5.** Which pairs of similar figures have a scale factor of 4 : 5? Circle all that apply.

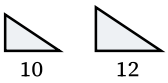
A.



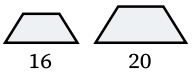

B.



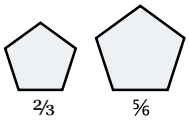

C.



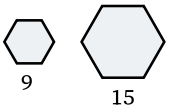

D.




E.

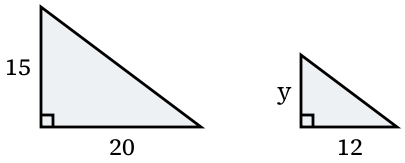



F.




A, D, E (8 : 10, 16 : 20 and ⅔ : ⅚ all simplify to 4 : 5; 12 : 16 = 3 : 4, 10 : 12 = 5 : 6, 9 : 15 = 3 : 5)

**6.** What is the value of $y$ in the similar figures?




A. 7

B. 8

C. 9

D. 16

C (20/12 = 15/y → 20y = 180 → y = 9)

**7. Two similar pentagons have a scale factor of 1 : 4. For numbers 7a – 7d, determine whether each statement about the pentagons is true or false.**

7a. Two corresponding sides could be 3 m and 12 m. **TRUE FALSE**

7b. The ratio of their perimeters is 1 : 16. **TRUE FALSE**

7c. Their areas could be 5 $in^2$ and 20 $in^2$. **TRUE FALSE**

7d. The ratio of their areas is 1 : 16. **TRUE FALSE**

**7a TRUE (3 : 12 = 1 : 4) · 7b FALSE (perimeters are 1 : 4) · 7c FALSE (5 : 20 = 1 : 4, but areas must be 1 : 16) · 7d TRUE ($1^2 : 4^2$)**

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## Part II – Constructed Response

***Solve each proportion.***

**1.** $\frac{7}{x} = \frac{21}{30}$

x = 10 (21x = 7 · 30 = 210; check 7/10 = 21/30)

**2.** $\frac{y}{16} = \frac{27}{36}$

y = 12 (36y = 16 · 27 = 432; check 12/16 = 3/4 = 27/36)

**3.** Gabe paid $11.94 for 6 bottles of juice. How much would 9 bottles of juice cost? Show all work necessary to justify your answer.

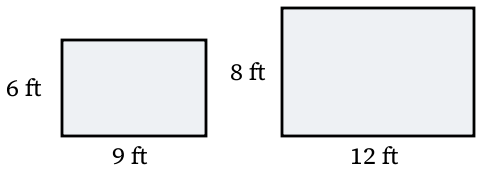
$17.91 (6/11.94 = 9/x → 6x = 107.46 → x = 17.91; or $1.99 per bottle × 9 = $17.91)

**4.** Rosa swam 12 laps in 15 minutes. Kim swam 18 laps in 24 minutes. Which person swam at a faster rate? Use mathematics to justify your answer.

Rosa. Rosa: 12 ÷ 15 = 0.8 lap per minute. Kim: 18 ÷ 24 = 0.75 lap per minute. 0.8 > 0.75. (Or: Rosa takes 1.25 min per lap, Kim takes about 1.33 min per lap.)

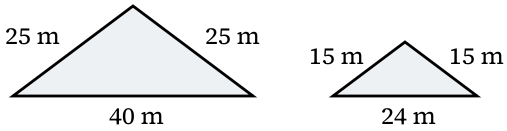
***Determine the scale factor for each pair of similar figures.***

**5.**



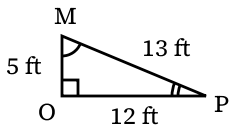

3 : 4 (9 : 12 = 6 : 8 = 3 : 4)

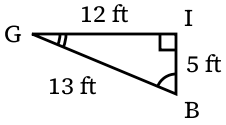
**6.**




5 : 3 (large to small: 25 : 15 = 40 : 24 = 5 : 3; small to large is 3 : 5)

**7.** Use the figures below to find the corresponding sides and corresponding angles to the ones identified. Matching arcs show congruent angles.




$\overline{MO}$ corresponds to _____ ∠M ≅ ∠_____

$\overline{OP}$ corresponds to _____ ∠O ≅ ∠_____

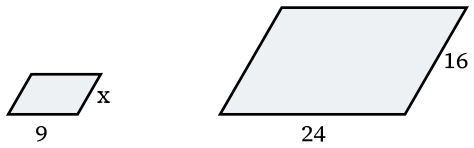
$\overline{MP}$ corresponds to _____ ∠P ≅ ∠_____

Are △MOP and △BIG congruent, similar, or neither? Explain your reasoning.

MO ↔ BI, OP ↔ IG, MP ↔ BG; ∠M ≅ ∠B, ∠O ≅ ∠I, ∠P ≅ ∠G. Congruent: the corresponding angles are congruent and every pair of corresponding sides is equal (5 = 5, 12 = 12, 13 = 13), so they are the same shape and the same size (scale factor 1 : 1). Congruent figures are also similar.

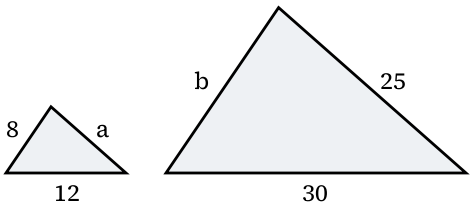
***Each pair of shapes below are similar. Solve for each variable. Show all work necessary to justify your answer.***

**8.**



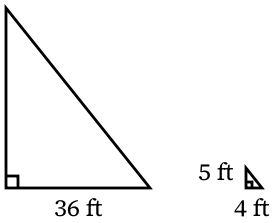

x = 6 (9/24 = x/16 → 24x = 144 → x = 6)

**9.**



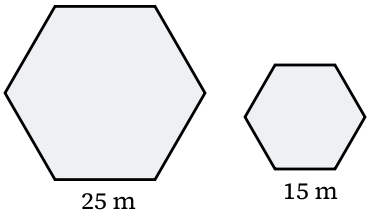

Scale factor 12 : 30 = 2 : 5. a: 12/30 = a/25 → 30a = 300 → a = 10. b: 12/30 = 8/b → 12b = 240 → b = 20.

**10.** Leo wanted to know the height of an apartment building on his street. One morning, he measured the building's shadow along the ground to be 36 feet long. At the same time, he measured his own shadow to be 4 feet long. Leo is 5 feet tall. Find the height of the building. Use mathematics to justify your answer.




45 feet (h/5 = 36/4 → 4h = 180 → h = 45; the building's shadow is 9 times Leo's, so it is 9 × 5 = 45 ft tall)

**11.** Use the similar regular hexagons below.




a. Find their scale factor (small to large).

3 : 5 (15 : 25)

b. Find the ratio of their perimeters.

3 : 5

c. Find the ratio of their areas.

9 : 25 ($3^2 : 5^2$)

d. Suppose the smaller hexagon has a perimeter of 90 m. What is the perimeter of the larger hexagon? Show all work necessary to justify your answer.

150 m (3/5 = 90/x → 3x = 450 → x = 150; check 6 × 25 = 150)

**12.** Two similar triangles have perimeters of 14 meters and 35 meters. The smaller triangle has an area of 12 $m^2$. Ava used this information to find the area of the larger triangle. Unfortunately, she made an error in her work. Explain the error and then find the correct area of the larger triangle.

> Scale factor is 14 : 35, which is 2 : 5
>
> Ratio of their areas is $2 \cdot 2 : 5 \cdot 2$, which is 4 : 10
>
> Let $x$ = the area of the larger triangle
>
> $\frac{4}{10} = \frac{12}{x}$
>
> $x = 30$
>
> The area of the larger triangle is 30 $m^2$.

**Ava multiplied each number by 2 instead of squaring it. The area ratio is $2^2 : 5^2 = 4 : 25$. 4/25 = 12/x → 4x = 300 → x = 75. The larger area is 75 $m^2$.**

**13.** A model train is built with a scale of 1 inch : 4 feet. A real passenger car is 52 feet long. How long is the model of the passenger car? Show all work necessary to justify your answer.

**13 inches (1/4 = x/52 → 4x = 52 → x = 13)**

# Core Focus on Proportions & Probability

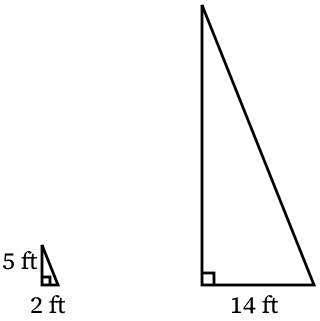
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## Part III – Problem Solving

Kayla is measuring a young oak tree in a park. She is 5 feet tall and her shadow along the ground has a length of 2 feet. At the same time, the oak tree casts a shadow along the ground that has a length of 14 feet. In a few years the tree will be 10 feet taller. How long would its shadow along the ground be then, at the same time of day?




**Tree now: h/5 = 14/2 → 2h = 70 → h = 35 ft. Ten feet taller: 35 + 10 = 45 ft. New shadow: 45/5 = s/2 → 5s = 90 → s = 18 ft. The shadow would be 18 feet long.**